Pediatrics High-Yield Review — Corrected Addendum 2026

A verified high-yield addendum to the uploaded *LEK Last Minute — Pediatria* notes — not a replacement edition of the 371-page source.

> **Medical-use notice.** I am an AI, not a medical professional—treat this as informed educational analysis, not a medical diagnosis or a treatment protocol. A qualified clinician should confirm anything consequential. Drug doses, resuscitation algorithms, vaccine schedules, and local referral pathways must be checked against the current institutional and national guideline before use.

Revision status: High-yield companion and errata edition. I reviewed selected high-risk and high-yield areas; I did **not** validate every statement, drug dose, or answer option in the 371-page source. **Last checked:** August 2026. **Jurisdiction note:** vaccine and treatment recommendations vary by country; schedule references below use current U.S. CDC/AAP source pages where stated.

Read this first — what was actually corrected

This repaired document is deliberately narrower and more transparent than the source PDF. It is designed for **teaching images and flashcard review**, where a wrong timing threshold, a stale algorithm, or an unlabelled old answer key can be memorized quickly.

Source domain	Status in this addendum	What was added or corrected
Newborn examination	**Reviewed high-yield**	Apgar framing; respiratory-score caveats; reflex timing; Babinski physiologic period; fontanelles
Development	**Reviewed high-yield**	cueing and milestone anchors; regression warning
Genetics	**Reviewed high-yield**	aneuploidy mechanisms; Down, Edwards, Patau, Turner, and Klinefelter quick atlas
Scoring and examination	**Reviewed high-yield**	Centor/McIsaac; Westley caveat; Barlow versus Ortolani; signs as clues
Resuscitation and respiratory care	**Targeted errata**	CDH ventilation wording; no routine early bicarbonate; bronchiolitis principles
Infection and immunization	**Targeted errata**	HSV delivery wording; current-schedule warning; infant infection caveat
Nephrology, GI, allergy, rheumatology	**Core pattern review**	UTI specimen caveat; nephrotic/nephritic patterns; IgA vasculitis; JIA versus growing pains
Drug doses, local algorithms, full vaccine tables, full oncology/endocrinology/metabolism sections	**Not fully reviewed**	Keep the original only as an index and verify against current local or national guidance

> **Rule for studying:** use a card or original-note statement as a memorization target only when it is labelled **Core**, **Caveat**, **Current check**, or **Erratum** in this addendum. Unlabelled source

> text remains unverified.

Teaching-atlas panel: timing, normal variation, and abnormal persistence

Corrections that were also added to the interactive flashcard deck

The study app now contains **380 cards**. Cards **359–380** are new verified review cards covering Babinski, Moro, rooting, sucking, palmar grasp, stepping, ATNR, fontanelles, Apgar, Centor/McIsaac, Barlow/Ortolani, chromosome notation, trisomies, HSV delivery, neonatal bicarbonate, CDH ventilation, measles infectiousness, vaccine-schedule caution, nephrotic/nephritic patterns, and sign interpretation. In the app, use the **Corrections** queue to review all caveated or invalid/outdated cards together.

How to use this edition

The original PDF is useful as a broad index, but it mixes translated textbook prose, memory aids, exam questions, and current-sensitive clinical statements. This edition keeps the high-yield pattern-recognition content and replaces or qualifies material that should not be memorized without verification.

Label	Meaning
Core	Stable examination concept suitable for active recall.
Caveat	Correct concept, but age, prematurity, local protocol, or rubric variation matters.
Current check	Reconfirm before clinical use because the recommendation can change.
Erratum	The source wording is misleading, incomplete, or incorrect and should not be memorized as written.

1. Neonate and Newborn Examination

Apgar score — Core

Apgar is a structured description of newborn transition, recorded at **1 and 5 minutes**. Each domain receives 0, 1, or 2 points.

Domain	0	1	2
Appearance	Pale or blue	Body pink, extremities blue	Entirely pink
Pulse	Absent	<100 beats/min	≥100 beats/min
Grimace	No response	Grimace	Cough, sneeze, or cry
Activity	Limp	Some flexion	Active movement / well flexed
Respiration	Absent	Slow or irregular	Good respirations / vigorous cry

Interpretation is commonly described as **7–10 reassuring**, **4–6 moderately abnormal**, and **0–3 low**, but the score is a snapshot, not a stand-alone diagnosis of asphyxia or a predictor of an individual child's long-term outcome. Resuscitation and effective ventilation take priority over waiting for a score.[1]

Neonatal respiratory distress — Caveat

Look first at the infant, not the score: respiratory rate, work of breathing, color/oxygenation, perfusion, tone, feeding ability, and response to support. A score can support serial assessment but does not replace clinical examination, pulse oximetry, blood gas assessment when indicated, or escalation of care.

Silverman–Andersen commonly grades upper-chest movement, lower-chest retractions, xiphoid retractions, nasal flaring, and expiratory grunting, with each domain scored 0–2. **Downes** versions commonly include cyanosis/oxygenation, respiratory rate, retractions, grunting, and air entry/wheeze. Downes cutoffs vary by version; do not import a threshold from one local chart into another.

Primitive reflexes — Caveat

The expected response and approximate integration window are more useful than a single memorized month. Prematurity, alertness, medication exposure, illness, asymmetry, and overall neurologic examination modify interpretation.

Reflex	Stimulus	Expected response	Approximate timing
Rooting	Stroke cheek or corner of mouth	Turns toward stimulus, opens mouth	Present by late gestation; usually integrates by about 4–6 months
Sucking	Touch palate/posterior tongue	Rhythmic suck	Appears around 32–35 weeks' gestation; coordination with swallowing and breathing matures toward term
Moro / startle	Sudden loss of support or startle	Abduction/extension, then flexion and cry	Present by late gestation; usually integrates by about 4–6 months
Palmar grasp	Finger placed in palm	Finger flexion/grasp	Usually integrates by about 4–6 months
Stepping	Hold upright with soles touching surface	Alternating step-like movements	Often fades by about 2 months; timing varies
Asymmetric tonic neck / fencing	Turn head to one side	Ipsilateral extension and contralateral flexion	Usually integrates by about 5–7 months
Babinski / infant plantar response	Stroke lateral sole	Great-toe dorsiflexion with fanning	Present from infancy; an extensor response is physiologic through approximately age 2 years. Persistence beyond about 2 years, marked asymmetry, or an abnormal neurologic context is concerning.

For LEK-style recall: **Babinski should be present in infancy, remain physiologic through roughly 1–2 years, and become concerning when it persists beyond about age 2, is clearly asymmetric, or occurs with other upper-motor-neuron signs.** An absent, markedly asymmetric, exaggerated, or persistently obligatory response is not a diagnosis by itself;

reassess the child's state, gestational age, tone, strength, cranial nerves, gait, and developmental trajectory.[2,17]

Fontanelles — Core + Caveat

The **anterior fontanelle** is the larger soft space near the front/top of the skull. The **posterior fontanelle** is smaller and lies near the back. A normal fontanelle is generally soft and flat when the infant is calm and upright.

Finding	High-yield interpretation
Posterior closure	Commonly within about 2 months, with normal variation
Anterior closure	Commonly around 7–18 months; variation is expected
Transient prominence while crying or lying down	May be physiologic if it returns to baseline when calm/upright
Persistent bulging when calm/upright	Consider raised intracranial pressure, meningitis/encephalitis, hydrocephalus, hemorrhage, or other urgent pathology
Sunken fontanelle	Dehydration is a common concern; assess hydration, feeding, urine output, perfusion, and illness severity
Early closure with abnormal head shape	Consider craniosynostosis and assess head growth and sutures
Delayed closure or unusually large fontanelle	Consider normal variation and conditions such as hypothyroidism, skeletal dysplasia, Down syndrome, or raised intracranial pressure; use the full examination

Closure timing should be documented as an approximate range, not a rigid cutoff. Head circumference trajectory and the infant's clinical state matter more than a single palpation.[3]

Responsive infant cueing — Core

Use the sequence **Observe** → **Interpret** → **Respond**. Engagement/readiness cues may include eye contact or looking toward the caregiver, relaxed face/body, reaching, vocalizing, smiling, and hands-to-mouth behavior that may signal hunger. Pause/reduce-stimulation cues may include gaze aversion, turning away, yawning, hiccups, finger splaying, fussing, stiffening, or color change. The correct response is to reduce stimulation, soften, pause, and observe again rather than forcing interaction.

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2. Developmental Anchors

Milestones are **age-linked anchors, not rigid diagnostic cutoffs**. Use surveillance, screening, hearing/vision assessment, neurologic examination, and longitudinal trajectory when concerns arise.

Approximate age	High-yield anchor
2 months	Social smile
4 months	Laughs/chuckles; holds head steady
6 months	Recognizes familiar people; increasing reciprocal interaction
9 months	Stranger reaction; sits without support
12 months	Waves bye-bye
18–24 months	Points, uses words, combines two words

Loss of a previously acquired skill is more concerning than an isolated mild delay and warrants prompt clinical assessment.

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3. Chromosomal Abnormalities and Genetics

Notation and mechanisms

- **Trisomy:** three copies of a chromosome or extra chromosome material.
- **Monosomy:** one copy rather than the usual pair.

- **Nondisjunction:** chromosome-separation error during cell division; a common mechanism of aneuploidy.
- **Translocation:** chromosome material is attached to another chromosome. A balanced translocation may be clinically silent in a carrier but can produce unbalanced material in offspring.
- **Mosaicism:** genetically different cell lines in the same individual; phenotype can vary.
- **Karyotype examples:** 47,XX,+21 and 47,XY,+21 describe an extra chromosome 21; 45,X is a common shorthand for Turner syndrome; 47,XXY is the classic Klinefelter notation.

High-yield syndrome table

Condition	Chromosomal pattern / mechanism	High-yield associations
Down syndrome	Usually trisomy 21; also translocation or mosaic forms	Hypotonia, characteristic facial features, developmental disability of variable degree, congenital heart disease, hearing/vision issues, thyroid disease, and increased risk of certain hematologic disorders
Edwards syndrome	Trisomy 18	Severe multisystem congenital anomalies, growth restriction, clenched hands with overlapping fingers, rocker-bottom feet, and significant developmental impairment
Patau syndrome	Trisomy 13	Severe multisystem anomalies; holoprosencephaly spectrum, cleft lip/palate, polydactyly, eye/brain abnormalities, and major developmental impairment
Turner syndrome	Monosomy X or mosaic/structural X abnormality; classically 45,X	Short stature, gonadal dysgenesis, delayed/absent puberty without treatment, webbed neck or lymphedema in some patients, coarctation/bicuspid aortic valve risk, and renal anomalies
Klinefelter syndrome	Usually 47,XXY; mosaic forms occur	Tall stature, small firm testes, hypergonadotropic hypogonadism, infertility, gynecomastia in some patients, and variable language/learning differences

Do not use phenotype alone to diagnose a chromosomal condition. Confirmatory testing, genetic counseling, and syndrome-specific surveillance are required. The uploaded PDF’s scattered references to Turner and trisomies are not a substitute for a complete genetics review.[4–8]

4. Scoring Systems and Examination Patterns

Centor / McIsaac score — Current check

The four clinical criteria are fever >38°C, tonsillar exudate or swelling, tender/swollen anterior cervical nodes, and absence of cough. Each contributes +1. McIsaac adds age adjustment: **+1 for ages 3–14, 0 for ages 15–44, and −1 for age ≥45**. The score supports risk stratification and testing decisions; it does not replace clinical assessment or local protocol. It is not a reason to prescribe antibiotics without appropriate evaluation.[9]

Westley croup score — Caveat

Common domains are level of consciousness, cyanosis, stridor, air entry, and retractions. A commonly used severity ladder is 0–2 mild, 3–5 moderate, 6–11 severe, and ≥12 impending respiratory failure, but institutional rubrics and management pathways vary. Always assess appearance, work of breathing, fatigue, hydration, and response to treatment; do not delay airway support while calculating a score.

Barlow versus Ortolani — Core

Maneuver	What it tests	Direction	Positive finding
Barlow	Whether a reduced hip is unstable and can be displaced	Flex hip, gently adduct, apply gentle posterior pressure	Palpable clunk/instability as the femoral head moves out
Ortolani	Whether a dislocated hip can be reduced	Flex hip, abduct, lift the greater trochanter anteriorly	Palpable clunk as the femoral head reduces

These maneuvers are most useful in young infants; sensitivity decreases after roughly 2–3 months. Never force the hip. Evaluate limited abduction, leg-length asymmetry, thigh-fold asymmetry, and gait findings according to age and local screening guidance.

High-yield signs are clues, not diagnoses

Finding	Classic association
Scaphoid abdomen	Congenital diaphragmatic hernia
Olive-shaped mass	Hypertrophic pyloric stenosis
Currant-jelly stool	Intussusception
Steeple sign	Croup
Thumb sign	Epiglottitis
Koplik spots	Measles
Palpable purpura	IgA vasculitis
Bulging fontanelle	Raised intracranial pressure / meningitis concern

Several signs have differentials and variable sensitivity. The full clinical picture, age, vital signs, and urgency always take priority.

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5. Emergency and Respiratory Principles

First approach

Use **airway, breathing, circulation, disability, exposure** as a structured first pass. Call for help early in a deteriorating child. Reassess after every intervention. PALS and neonatal-resuscitation algorithms are updated periodically; use the current course materials and local protocol rather than a static textbook table.[10–12]

Bronchiolitis — Current check

Typical bronchiolitis is a clinical diagnosis in infants and young children. Uncomplicated cases generally receive supportive care: oxygenation assessment, hydration/feeding support, nasal suction when needed, and escalation for apnea, exhaustion, severe work of breathing, hypoxemia, or inability to maintain hydration. Routine chest radiography, broad laboratory testing, antibiotics, systemic corticosteroids, and routine bronchodilator treatment are not appropriate for every uncomplicated case; local guideline details differ.[13]

Congenital diaphragmatic hernia — Erratum correction

If CDH is known or strongly suspected, avoid prolonged bag–valve–mask ventilation because gastric and bowel inflation can worsen lung compression. Prioritize expert airway management and gastric decompression according to neonatal-resuscitation and surgical protocols.[14]

Neonatal sodium bicarbonate — Erratum correction

The source PDF's statement that sodium bicarbonate should be administered at the beginning of resuscitation is unsafe as a general rule. Current neonatal-resuscitation guidance does **not** support routine sodium bicarbonate during initial newborn resuscitation; ventilation, compressions when indicated, epinephrine when indicated, and correction of reversible causes take priority. Any exceptional use is protocol- and specialist-dependent.[11,12]

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6. Infection and Immunization

Vaccine schedules — Current check

Do not copy an annual schedule into a timeless study guide. The CDC child/adolescent schedule page and its appendix should be checked for the current year, catch-up recommendations, medical indications, contraindications, and precautions. The 2025 schedule page includes updates and links to age-based, catch-up, medical-indication, and contraindication tables.[15]

General exam principle: a severe allergic reaction to a previous dose or a component is a contraindication to that vaccine; moderate or severe acute illness is commonly a precaution rather than an automatic permanent contraindication. Live-vaccine restrictions depend on immune status, pregnancy, and specific products. Always use the current schedule for the patient's country and clinical context.

HSV in pregnancy — Erratum correction

The source PDF says that HSV-2 infection is a contraindication to cesarean delivery and repeats that cesarean should be avoided. That wording is incorrect. Cesarean delivery is generally recommended when active genital lesions or prodromal symptoms are present at labor because it can reduce neonatal transmission risk; management also depends on primary versus recurrent infection, suppressive therapy, membrane status, and obstetric guidance.[16]

Meningitis and infant infection

Infants may lack classic meningeal signs. Irritability, poor feeding, lethargy, temperature instability, apnea, seizures, vomiting, bulging fontanelle, or abnormal perfusion require age-appropriate urgent assessment. Do not use a single sign or a single CSF value in isolation.

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7. Nephrology, Gastroenterology, Allergy, and Rheumatology

Pediatric UTI / pyelonephritis

Fever without a source in an infant can represent UTI. Diagnosis requires an appropriately collected urine specimen; bag urine is not adequate for culture confirmation. Interpret urinalysis and culture with age, symptoms, collection method, prior antibiotics, and local resistance patterns. Imaging and prophylaxis decisions depend on age, atypical/recurrent infection, renal findings, and local guideline.

Nephrotic versus nephritic pattern

Pattern	High-yield features
Nephrotic	Heavy protein loss, hypoalbuminemia, edema, and hyperlipidemia
Nephritic	Hematuria, hypertension, reduced kidney function, and variable proteinuria

Do not memorize a single antibiotic, steroid, or biopsy pathway without checking the current syndrome-specific guideline.

IgA vasculitis

Think of palpable purpura, usually on dependent areas, with abdominal pain, arthralgia/arthritis, and possible renal involvement. Monitor blood pressure and urinalysis over time; a benign initial appearance does not eliminate the need for follow-up.

JIA versus growing pains

Growing pains are typically intermittent, bilateral, non-articular limb pains with a normal examination and no persistent morning stiffness or functional loss. JIA is suggested by objective joint swelling, warmth, limited range of motion, persistent morning stiffness, limp, or systemic features. Persistent, unilateral, nocturnal, or function-limiting pain requires assessment rather than reassurance by label.

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8. Errata from the Uploaded Source

The following items should be marked in the original PDF rather than memorized as written.

Source issue	Corrected study wording
HSV-2 is described as a contraindication to cesarean section	Active genital lesions or prodrome at labor generally support cesarean delivery to reduce neonatal transmission risk; follow obstetric guidance
Sodium bicarbonate is presented as a first-choice drug at the beginning of neonatal resuscitation	Routine bicarbonate is not part of initial neonatal resuscitation; follow current neonatal-resuscitation guidance
Mouth-to-mouth ventilation is broadly described as contraindicated because of esophageal rupture	In known/suspected CDH, avoid prolonged bag–mask ventilation because gastric insufflation worsens lung compression; emergency ventilation decisions depend on the clinical scenario
Fontanelle wording is incomplete or anatomically unclear	Use anterior/posterior anatomy, approximate closure windows, and calm/upright bulging versus dehydration-related sunken findings
Vaccine schedules and contraindications are presented as static facts	Replace with a dated link to the current national schedule and its contraindication/precaution appendix
Isolated physical signs are treated as direct diagnoses	Present each as a clue with differentials, age context, sensitivity limitations, and urgency caveats
Memory aids and translated phrases are mixed with verified clinical recommendations	Separate mnemonic material from evidence-based recommendations and label local-protocol-dependent content

9. Source and Update Register

The following references were used to correct or qualify the highest-risk sections. They should be rechecked whenever the guide is reused for a new exam cycle.

1. Apgar score — Merck Manual Professional Edition
2. Primitive Reflexes: Comprehensive Neurological Assessment — NCBI Bookshelf
3. Anterior and Posterior Fontanelle Closures — Children’s Hospital Colorado

4. Down Syndrome — MedlinePlus Genetics
5. Trisomy 13 — MedlinePlus Genetics
6. Trisomy 18 — MedlinePlus Genetics
7. Klinefelter Syndrome — MedlinePlus Genetics
8. Turner Syndrome — MedlinePlus
9. Centor Score / Modified McIsaac — MDCalc reference
10. 2025 AHA Pediatric Advanced Life Support Guidelines
11. 2025 AHA Neonatal Resuscitation Guidelines
12. AAP Neonatal Resuscitation Guidelines
13. UCSF Bronchiolitis Guidelines
14. Diagnosis and management of congenital diaphragmatic hernia — clinical practice guideline
15. CDC Child and Adolescent Immunization Schedule
16. CDC Herpes STI Treatment Guidelines
17. MedlinePlus — Babinski reflex

Review reminder: This document is a corrected high-yield companion, not a replacement for current local protocols, specialist guidance, or formal resuscitation training.